

Deviation Reporting for Distance and Time at EOD

Last modified: July 3rd '24

Problem Alignment: Team Managers using DispatchOne's routing capability don't get a visibility around the efficiency of a dispatch plan and there is no comparison between the planned Vs actual performance of distance traveled and time consumed to complete a particular dispatch. Due to this, an objective assessment of driver's performance and dispatch plans efficiency becomes difficult to establish.

Hence, this document captures the solution for calculating and displaying the deviation between planned and actual performance of a dispatch at the time of EOD in terms of :

- Time taken by the driver for completing the dispatch
- Distance traveled by the driver for completing the dispatch

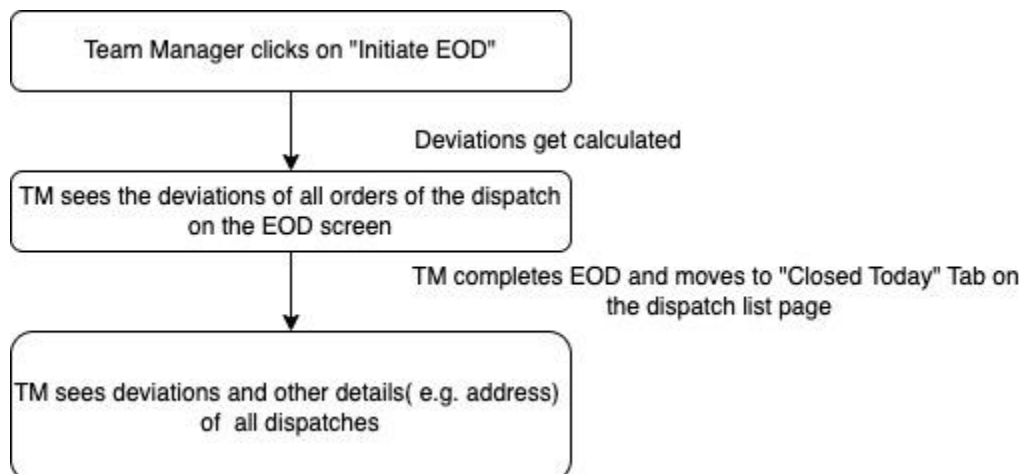
Scope:

- This solution will be available to all clients who have subscribed to the Routing Add-on app irrespective of their geographical location and without any dependency on any client config.
- Deviation will be visible only for closed dispatches and will be calculated only at EOD

Out of scope:

- No realtime computation of deviation before EOD
- There will not be any deviation reporting for any manual dispatch or for any automated dispatch with manual intervention.

Expected User behavior:



Solution:

Step 1: Calculation of Time and Distance:

1. Store “Planned Distance” and “Planned time” calculated by Routing engine against every dispatch at the time of publishing a dispatch.
 - a. If there is any manual intervention, system will not have any “Planned metrics” and no deviation will be calculated for such dispatches
2. Store total distance traveled by a driver in a dispatch as received from MTS. If the flag for entering manual odometer reading is enabled for the particular client then the reading entered manually by the Driver will override the MTS calculated reading.
3. Calculate actual time taken by a driver in a dispatch as follows:
 - a. “EOD Completion Time” - “Start Dispatch time” for tenants where EOD is enabled
 - b. “Dispatch Closure Time” - “Start Dispatch time” for tenants where EOD is not enabled
4. Calculate Order completion time
 - a. Time when drop objective is met for a FWD order and pick objective is met for a RVP order. This objective can be met either by explicit input from the driver or automatically by the system based on some logic like “exit from geofence”[Times of India future use case]. This means there can be multiple orders with the exact same time stamp if the backend logic gets triggered at the same time.
 - b. If a driver does not mark an order as delivered/picked in the FE App and no automatic logic is applicable for the particular tenant then “EOD Completion Time” will be taken as default completion time. This means there can be multiple orders with the exact same time stamp.

Step 2: Providing visibility of Time and Distance information on UI

EOD Screen (Automatic Plan): TM sees Actual distance, Actual Time and deviations

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End of Dispatch

Pembroke Pines-170523-143954 Ongoing

Total Orders	Not Started	In Progress	Completed	Failed	Expected Cash	Cash in Hand	Distance	Time
200	0	0	148	52	₹ 500	₹ 200	80 KM	1 HR 56 MIN
1221 Shipments	0 Shipments	0 Shipments	907 Shipments	314 Shipments			Travelled 10 km more	Took 40 min more

Update Order Status

All 200
 Completed 148
 Failed 52
 Ongoing 0

Q

Search by Order ID, Shipment ID

Order Type

Deviation

Status

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⋮

☐ ORDER ID	SHIPMENT ID	NO. OF SHIPMENTS	STATUS	PAYMENT TYPE	COMPLETED AT	REASON
☐ 6382...7293 FWD	9227...6272 +6	7	Completed	Prepaid	9:00 AM 20 min late	Delivered to consignee
☐ 6382...4234 FWD	8003...3994 +9	10	Completed	Prepaid	9:20 AM 20 min late	Delivered to consignee
☐ 6382...1223 REV	3949...4523 +3	4	Completed	Prepaid	9:30 AM 5 min early	Return completed
☐ 6382...3022 REV	8204...2433 +4	1	Completed	Prepaid	9:40 AM	Return completed

Deviation is to be calculated when distance has a high and medium confidence score. The color coding (Green for High, Orange for Medium and Red for Low) for the same is to be shown on the GPS/Mobile icon.

EOD Screen (Manual Plan): Show Actual distance and Actual Time

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End of Dispatch

Pembroke Pines-170523-143954 Ongoing

Total Orders	Not Started	In Progress	Completed	Failed	Expected Cash	Cash in Hand	Distance	Time
200	0	0	148	52	₹ 500	₹ 200	80 KM	1 HR 56 MIN
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Update Order Status

All 200
 Completed 148
 Failed 52
 Ongoing 0

Q

Search by Order ID, Shipment ID

Order Type

Deviation

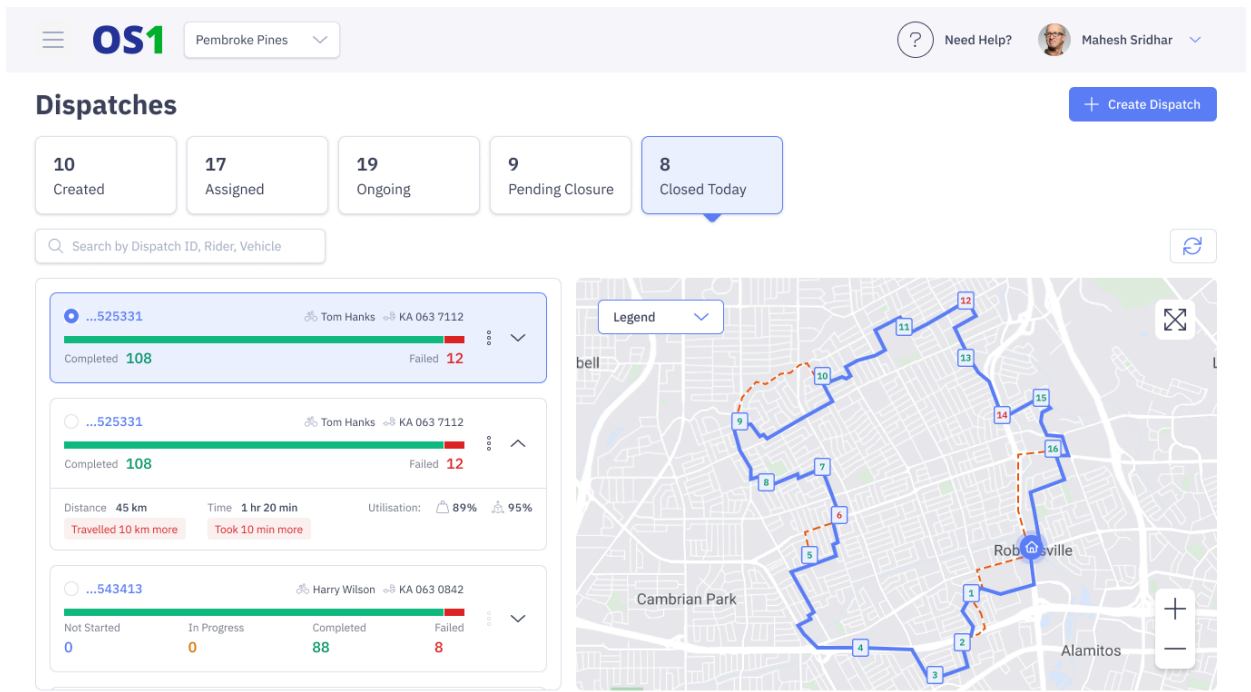
Status

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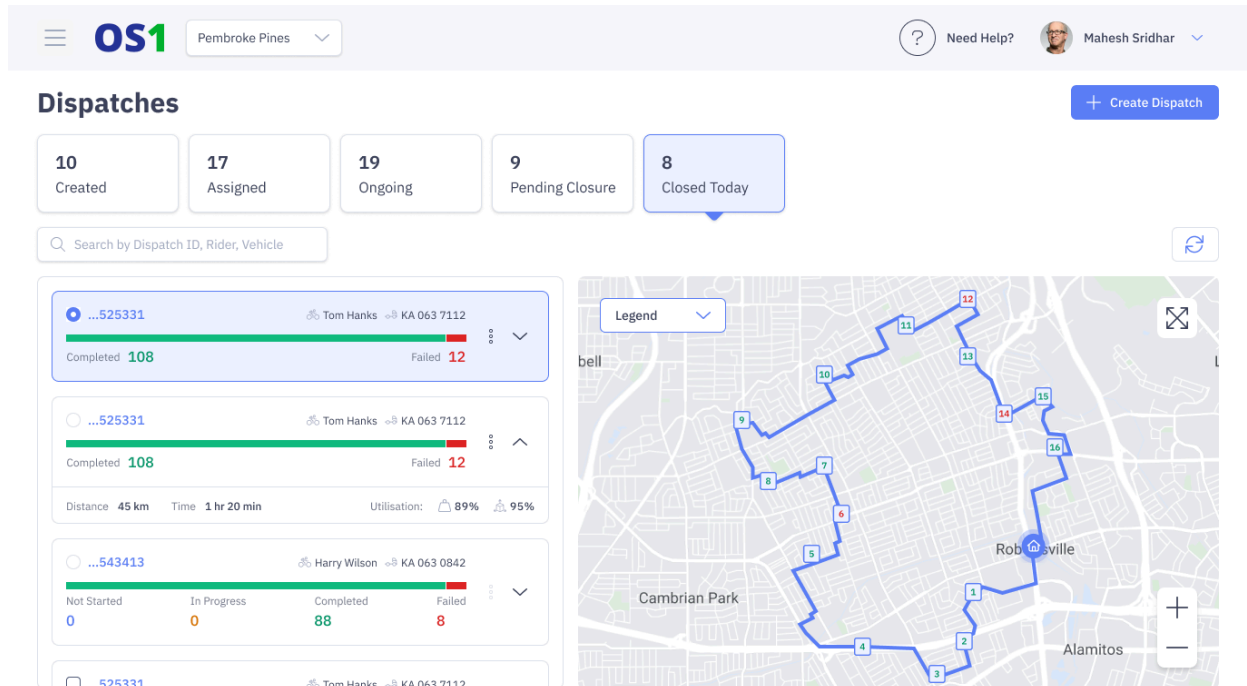
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☐ ORDER ID	SHIPMENT ID	NO. OF SHIPMENTS	STATUS	PAYMENT TYPE	COMPLETED AT	REASON
☐ 6382...7293 FWD	9227...6272 +6	7	Completed	Prepaid	9:00 AM	Delivered to consignee
☐ 6382...4234 FWD	8003...3994 +9	10	Completed	Prepaid	9:20 AM	Delivered to consignee
☐ 6382...1223 REV	3949...4523 +3	4	Completed	Prepaid	9:30 AM	Return completed
☐ 6382...3022 REV	8204...2433 +4	1	Completed	Prepaid	9:40 AM	Return completed

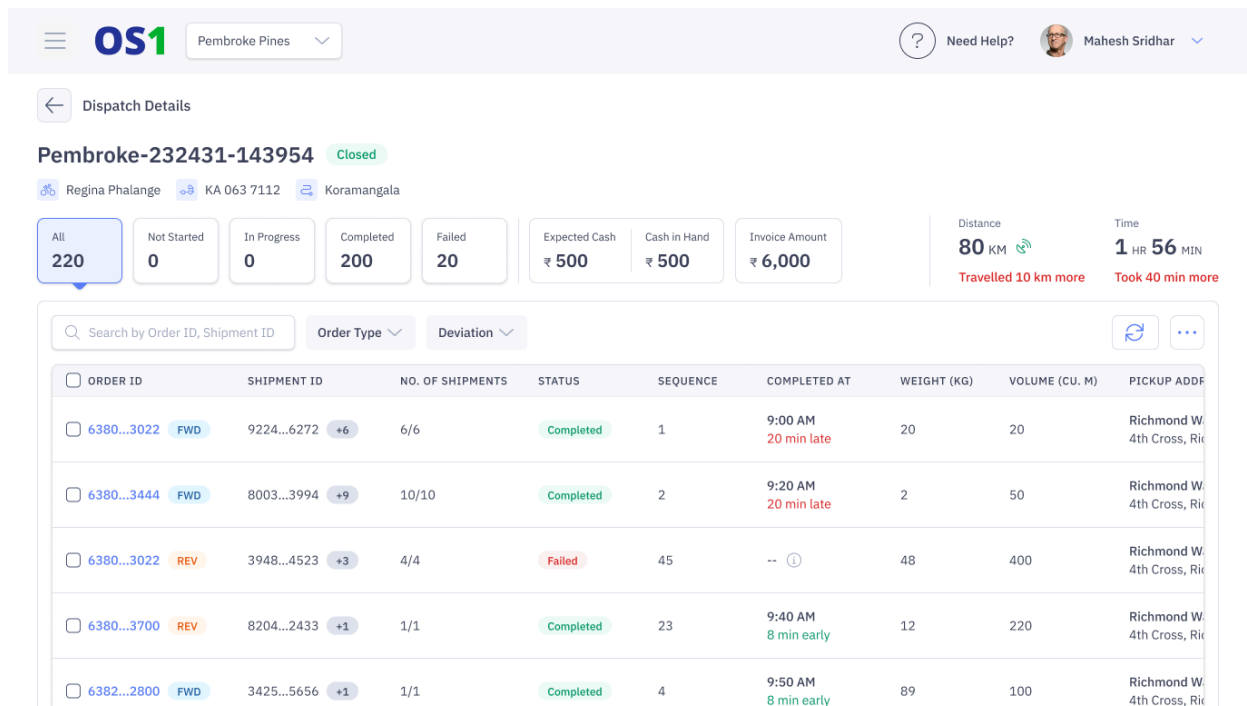
Dispatch List page (Automatic Plan): Show Actual distance, Actual time and deviations when a dispatch is in **“Closed” state**. TM will also be able to select on any dispatch and see the planned and actual path on the map on UI. Only one dispatch can be selected at a time and traces corresponding to only 1 plan will be visible on the UI for closed dispatches. [Actual Distance is already visible on UI, introduce a column for Actual Time as well- this will increase one column in the table]



Dispatch List page (Manual Plan): Show Actual distance, Actual time when a dispatch is in **“Closed” state**



Dispatch Details page (Automatic Plan): Show Actual distance, Actual Time and deviations



Dispatch Details page (Manual Plan- No planned value): Show Actual distance and Actual Time

OS1

Pembroke Pines

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Need Help?

Maresh Sridhar

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Dispatch Details

Pembroke-232431-143954

Closed

Regina Phalange

KA 063 7112

Koramangala

All

Not Started

In Progress

Completed

Failed

Expected Cash

Cash in Hand

Invoice Amount

Distance

Time

220

0

0

200

20

₹ 500

₹ 500

₹ 6,000

80 KM

1 HR 56 MIN

Search by Order ID, Shipment ID

Order Type

ORDER ID	SHIPMENT ID	NO. OF SHIPMENTS	STATUS	SEQUENCE	COMPLETED AT	WEIGHT (KG)	VOLUME (CU. M)	PICKUP ADDR
☐ 6380...3022	FWD 9224...6272 +6	6/6	Completed	1	9:00 AM	20	20	Richmond W 4th Cross, Ri
☐ 6380...3444	FWD 8003...3994 +9	10/10	Completed	2	9:20 AM	2	50	Richmond W 4th Cross, Ri
☐ 6380...3022	REV 3948...4523 +3	4/4	Failed	45	-- ①	48	400	Richmond W 4th Cross, Ri
☐ 6380...3700	REV 8204...2433 +1	1/1	Completed	23	9:40 AM	12	220	Richmond W 4th Cross, Ri
☐ 6382...2800	FWD 3425...5656 +1	1/1	Completed	4	9:50 AM	89	100	Richmond W 4th Cross, Ri

Order Details page [No Change here] : Don't want to highlight any deviation at the order details page because

- An order might deviate from the planned slot but can still get delivered on time and not get delayed.
- Deviation is a property of a dispatch and not individual orders so I want to keep that only in Dispatch views
- An order can be a part of many dispatch plans so comparison between multiple sets will be required to be done

Figma:

<https://www.figma.com/design/DWmDtliNXekrqi2VtjlPlc/Dispatch-FPA?node-id=8532-334379&t=T8nGlCA2KxLfOezj-11>